

XRComm FCS Wideband Platform & Playback

Quick Start Guide

Version 1.4.3 · June 2026

1. Introduction

The XRComm FCS Wideband Platform is a high-performance RF signal capture and distribution system designed for 8-channel, dual-port receive operations. It accepts raw IQ data from two HSP ports and makes that data simultaneously available to logging, DSP, and IQ application consumers.

The platform also includes the XRComm Universal Playback engine, which streams raw I/Q signal data from a binary waveform file directly to a network interface at line rate (100 Gbps).

2. Platform Components

Component	What It Does
<code>xrcomm_server</code>	The platform driver executable. Receives IQ packets, demultiplexes channels, and dispatches data to consumers simultaneously.
<code>xrcomm_grpc_ctrl</code>	The gRPC control server. Runs alongside the platform driver and exposes a network API for enabling/disabling modes and telemetry.
<code>xrcomm_playback</code>	Waveform playback tool. Transmits pre-recorded .bin waveform files back onto the network at the configured sample rate.
<code>xrcomm_nvme_loader</code>	Offline NVMe tool. Loads pre-recorded waveform files directly onto the NVMe drive so the platform can stream them without a network source.

3. Full Topology Block Diagram

[image]

4. Running the Platform Driver

Navigate to the platform driver directory and run the server as root:

```
cd XRComm_FCS_wideband_platform/XRComm_platform_drivers/  
sudo ./bin/xrcomm_server
```

The server keeps the gRPC server internally to handle all RPC commands. Users should only interface with the platform via the gRPC Client running on the Client Computer.

5. Controlling the Platform over gRPC

The gRPC Client should be run on the **Client Computer**.

Endpoint Default: 127.0.0.1:50051. Use --host and --port flags to override.

RPC Reference

RPC	Request	Purpose
SetDspMode	BoolRequest	Enable/disable DSP mode (FULL_PACKET_MODE).
SetLogging	BoolRequest	Enable/disable NVMe trigger-gated IQ capture. Acts as start .
SetIpMode	BoolRequest	Enable/disable per-channel IQ dispatch to applications.
SetReadCapture	FloatRequest	Set the duration of the read capture in seconds.
SetReadRate	FloatRequest	Set the sample_rate in Hz.
GetStats	Empty	Combined pipeline counters and telemetry.
WatchStatus	Empty	1 Hz live status stream.
Shutdown	Empty	Gracefully stop the platform pipeline.

Appendix A: Environment Prerequisites

Supported Operating System: Ubuntu 22.04 LTS (64-bit) or Ubuntu 24.04 LTS.

Required Dependencies

Package	Version	Installation
DPDK	23.11.1	apt install dpdk libdpdk-dev
SPDK	24.01	Build from source
gRPC (C++ runtime)	1.62.0	apt install libgrpc++-dev libprotobuf-dev
Python 3 + grpcio	3.10+	pip3 install grpcio grpcio-tools
CMake	3.16+	apt install cmake
GCC	11+	apt install build-essential

Hugepages The platform driver requires 2 MB hugepages pre-allocated before launch.

```
echo 4096 | sudo tee /sys/kernel/mm/hugepages/hugepages-2048kB/nr_hugep
sudo mkdir -p /dev/hugepages
sudo mount -t hugetlbfs nodev /dev/hugepages
```

NIC Binding Both receive NIC ports must be bound to the VFIO-PCI driver before starting the platform.

```
sudo modprobe vfio-pci
sudo dpdk-devbind.py --bind=vfio-pci 0000:xx:00.0 0000:xx:00.1
```